

Enthalpy of Vaporization of Water

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PURPOSE

The purpose of this investigation is to determine the enthalpy of vaporization (ΔH_{vap}) of a solvent by measuring its boiling point as a function of pressure. A linear relationship between $\ln(P)$ and $\frac{1}{T}$ can be established through the use of the Clausius-Clapeyron equation. The slope of this line is the ΔH_{vap} of the solvent, generated by our experimental data. This slope can be used to calculate the vapor pressure of water at room temperature (25°C), which can be compared to the literature value to verify the results of the experiment.

THEORY

The tendency of a liquid to vaporize is quantified by its vapor pressure, which increases with temperature and is the pressure exerted by a vapor with its condensed phase (solid or liquid) in a closed system. The boiling point of a liquid is when the vapor pressure equals the surrounding atmospheric pressure. At atmospheric pressure (1 atm), this would be the normal boiling point. Liquids can evaporate at temperatures below their boiling point, producing a vapor above the liquid's surface.

A Liquid-vapor equilibrium is when both liquid and vapor are present, and this begins with the requirement that the Gibbs free energies of the gas and liquid phases are equal. If $\Delta G_g = \Delta G_l$ and then the second law of thermodynamics is applied for a constant-temperature and pressure environment, then $\Delta S = \frac{q}{T}$ and $q = \Delta H$.

If these expressions are substituted into the phase equilibrium expression, this yields the

Clapeyron Equation: $\frac{dP}{dT} = \frac{\Delta H_{vap}}{(T^*\Delta V)}$. Since the molar volume of the gas is much greater than the

molar volume of the liquid, it can be simplified that the $\Delta V = \text{molar volume of the Vapor}$. If the vapor is then treated as an ideal gas, $PV = nRT$ can be substituted to yield

Equation 1 (Clausius-Clapeyron Equation):

$$\frac{dP}{P} = \left(\frac{\Delta H_{vap}}{R} \right) * \frac{dT}{T^2}$$

This equation requires integration, but assuming the ΔH_{vap} is independent of temperature, which in non-extreme cases it is, two useful forms of the equation can be integrated,

Equation 2 (Integrated Form of Clausius-Clapeyron Equation):

$$\ln\left(\frac{P_2}{P_1}\right) = \frac{-\Delta H_{vap}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Equation 3 (Linear Form of Clausius-Clapeyron Equation):

$$\ln(P) = \left(\frac{-\Delta H_{vap}}{R} \right) \left(\frac{1}{T} \right) + B$$

Equation 3 is in the linear form $y = mx + b$, so a graph of $\ln(P)$ versus $\frac{1}{T}$ will be a linear equation with a slope of $\frac{-\Delta H_{vap}}{R}$. This form of the Clausius-Clapeyron equation enables the determination of the enthalpy of vaporization from the pressure and temperature data collected in this experiment.

EXPERIMENTAL

Equipment: Vacuum line and adjustable stopcock, reflux column, heating mantle, round-bottom flask, water, mercury barometer, pressure gauge, thermometer.

Experimental Procedure:

Atmospheric pressure was recorded using the barometer (771.5 mmHg). The glassware connections were verified to be secure, and the stopcock was tested by venting the system before any heating began.

The system was opened to the atmosphere, and the heating mantle was activated. Once the water began to boil, the pressure and temperature were recorded and remained stable for a minute. This was recorded as the first data point.

The stopcock was then rotated to connect the system to the vacuum line. The pressure was reduced incrementally by opening the vacuum, then the stopcock was closed to keep the system at each target pressure. Data was collected at approximate absolute pressures of 660, 615, 563, and 502 mmHg. At each pressure, the heating continued until a stable boiling condition was reached. A stable boiling condition was when the temperature and pressure remained constant for a minute. A final trial was recorded when the pressure was returned to atmospheric. After all the trials, the system was then vented to atmospheric pressure, the heating mantle was turned off, and the flask was allowed to cool to room temperature before disposal.

DATA

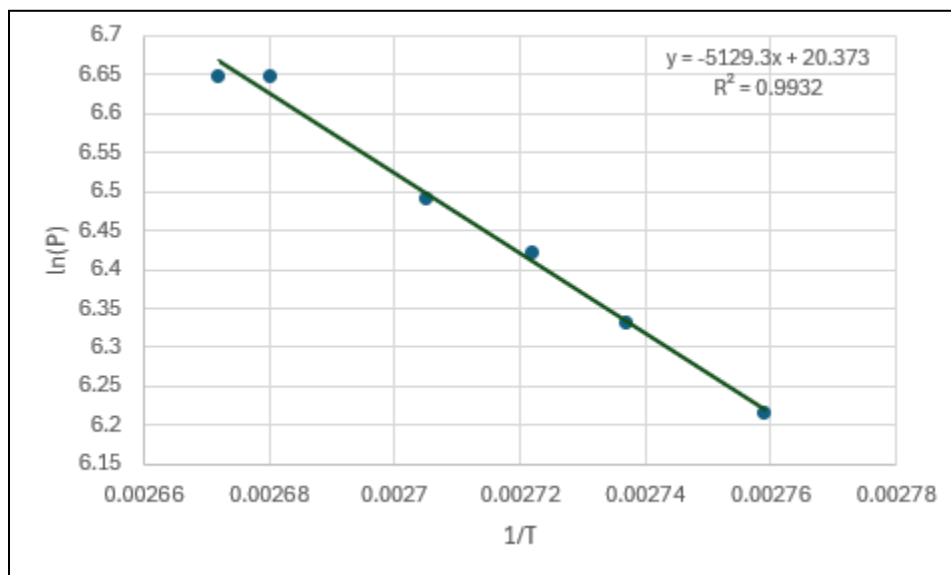
Atmospheric pressure was measured at 771.5 mmHg using the barometer. The absolute pressure was calculated by adding the gauge pressure (negative) to the atmospheric pressure. The temperature had to be converted from Celsius to Kelvin in order to use the Clausius-Clapeyron equation. The natural log of the absolute pressure and the inverse of the temperature were calculated and reported in Table 1.

Table 1: Experimental Data for Water

Temp (°C)	Gauge P (mmHg)	Absolute P (mmHg)	Temp (K)	1/T (K^{-1})	ln(P)
100.000	0	771.5	373.150	0.002680	6.6483
96.555	-112	659.5	369.705	0.002705	6.4915
94.225	-157	614.5	367.375	0.002722	6.4208
92.229	-209	562.5	365.379	0.002737	6.3324
89.332	-270	501.5	362.482	0.002759	6.2176
101.044	0	771.5	374.194	0.002672	6.6483

The data from **Table 1** were used to construct an $\ln(P)$ versus $1/T$ plot. The resulting graph is below:

Graph 1: $\ln(P)$ vs. $1/T$ (Clausius-Clapeyron Plot)



RESULTS

Question 1 - Tabulated Data with Units: Reported above in **Table 1**.

Question 2 - Enthalpy of Vaporization from Graph Slope: The slope of the $\ln(P)$ vs. $1/T$ plot was determined by linear regression to be -5129.3. Using the relationship from **Equation 3**:

$$-\Delta H_{vap} = -\text{slope} * R = -(-5129.3K) * 8.314J * mol^{-1} * K^{-1} = 42645J/mol$$

$$42,645 J/mol = 42.86 kJ/mol$$

Question 3 - Vapor Pressure at 25°C: Utilizing **Equation 2**, the vapor pressure at 25°C (298.15K) was estimated using the atmospheric trial ($P_1 = 771.5 mmHg$, $T_1 = 373.15K$):

$$\ln\left(\frac{P_2}{771.5}\right) = -\left(\frac{42645}{8.314}\right) * \left(\frac{1}{298.15} - \frac{1}{373.15}\right)$$

$$P_2 = 24.30 mmHg$$

Question 4 - Comparison to Literature Values: The experimental results can be compared with the literature values in **Table 2** below.

Table 2: Comparison of Experimental and Literature Values

Variable	Experimental Value	Literature Value
ΔH_{vap} (kJ/mol)	42.86	40.65
Vapor Pressure at 25°C (mmHg)	24.3	23.8
Normal Boiling Point (100°C)	100	100

The experimental ΔH_{vap} of 42.86 kJ/mol compares accurately to the literature value of 40.65 kJ/mol. These values yielded a percent error of 5.43%. The calculated vapor pressure of 25°C of 24.3 mmHg is within 2.10%. This indicates a strong agreement between the experimental and the literature values for the vapor pressure of water.

The small overestimation of the ΔH_{vap} value is likely an indication of systematic error present in the system. The most likely source is heat loss from the refluxing column to the surroundings. If the column is losing heat, the liquid would have to be heated to a higher temperature than it otherwise would need to be, inflating the ΔH_{vap} . There is also likely a pressure drift between the moment the vacuum line is closed and when the boiling temperature stabilizes. The atmospheric trials showed a slight deviation, which is another piece of evidence that the pressure or temperature was leaking or deviating from the read value, depending on the error.

REFERENCES

Boundless (now Lumen Learning). "2.13: Water – Heat of Vaporization." *Biology LibreTexts*, last updated November 23, 2024.

[https://bio.libretexts.org/Bookshelves/Introductory_and_General_Biology/General_Biology_\(Boundless\)/02%3A_The_Chemical_Foundation_of_Life/2.13%3A_Water_-_Heat_of_Vaporization](https://bio.libretexts.org/Bookshelves/Introductory_and_General_Biology/General_Biology_(Boundless)/02%3A_The_Chemical_Foundation_of_Life/2.13%3A_Water_-_Heat_of_Vaporization)

East Carolina University, Department of Chemistry. (2022). Physical Chemistry I Laboratory CHEM 3951 [Laboratory manual].

University of Oxford, Physical and Theoretical Chemistry Laboratory. "VR Chemistry: Solutions 3 — Vapor Pressure of Water." *vrchemistry.chem.ox.ac.uk*.

<https://vrchemistry.chem.ox.ac.uk/potential/text/solutions3.htm>